

POC REQUIREMENTS DOCUMENT

Project Name: **Capacity & Utilization Intelligence Agent (CUIA)**

Project Type: Proof of Concept

Owner: Noel Mathews, Priyatham Sai Kethavarapu

Objective:

This Proof of Concept demonstrates how engineering workforce data can be transformed into meaningful workforce insights using deterministic analytics and an AI-powered Copilot. It focuses on showcasing the core analytics engine, interactive dashboards, and AI-assisted explanations using a simulated engineering organization, while keeping the implementation simple and focused on validating the core concept.

Tech Stack:

- Frontend: React, Tailwind CSS, Recharts
- Backend: FastAPI, Python
- Analytics: Pandas, NumPy (Deterministic Analytics)
- AI: LangGraph, Google Gemini

In Scope:

- Simulated engineering dataset representing Delivery Managers and Engineers.
- Deterministic workforce analytics, including utilization, workload, productivity, estimation accuracy, and team health.
- Rule-based recommendations generated from workforce analytics.
- Leadership and Delivery Manager dashboards with interactive visualizations and key insights.
- AI Copilot for natural language queries, explanations, and analytics summaries.
- Modular application architecture designed for future expansion.
- Generate downloadable daily, weekly, and monthly workforce reports.

Out of Scope:

- Authentication and Role-Based Access Control (RBAC).
- Live integrations with Jira or other external systems.
- CSV imports, leave management, and skills management.
- User administration, system configuration, and audit logging.
- Background synchronization, scheduled jobs, and notification services.
- Multi-tenancy, high availability, CI/CD pipelines, and production deployment.
- Mobile applications and enterprise-grade security features.
- AI-driven analytics, autonomous decision-making, or data modification.

FUNCTIONAL REQUIREMENTS:

- **FR-1:** Load a predefined dataset representing a simulated engineering organization.
- **FR-2:** Calculate workforce analytics, including utilization, workload, productivity, estimation accuracy, and team health.
- **FR-3:** Generate rule-based recommendations from the calculated analytics.
- **FR-4:** Provide a Leadership Dashboard with organization-wide insights, KPIs, and team comparisons.
- **FR-5:** Provide a Delivery Manager Dashboard with team and engineer-level analytics.
- **FR-6:** Visualize workforce insights through interactive charts, KPI cards, and summary views.
- **FR-7:** Provide an AI Copilot that answers natural language queries using pre-calculated analytics.
- **FR-8:** Ensure the AI Copilot only explains analytics and recommendations without performing calculations or modifying data.
- **FR-9:** Support switching between Leadership and Delivery Manager views for demonstration purposes.
- **FR-10:** Generate downloadable daily, weekly, and monthly workforce reports summarizing analytics, KPIs, and recommendations.

NON-FUNCTIONAL REQUIREMENTS:

- **Simplicity:** The solution should focus on demonstrating the core concept with a simple and maintainable implementation.
- **Deterministic Analytics:** All workforce metrics and recommendations should be generated using deterministic business logic.
- **AI Boundaries:** The AI Copilot should only explain and summarize existing analytics and must not calculate metrics or modify data.
- **Performance:** Dashboards and report generation should provide fast access to precomputed analytics suitable for live demonstrations.
- **Architecture:** The application should follow a modular architecture with clear separation between the Analytics Engine, Recommendation Engine, AI Copilot, and Dashboard to support future enhancements.